

Formative R activities: Health Data Analysis

1. Objective

This explanatory analysis covers statistical activities conducted in R using the provided **Health Data** set. The dataset contains 210 observations and 17 variables, including age, sex, religion, occupation, income, and medical conditions such as blood pressure, peptic ulcer, and diabetes; pre- and post-test data; and admission and discharge data.

The activities focused on data exploration, data visualisation, hypothesis testing, and building a linear regression model.

2. Dataset Exploration

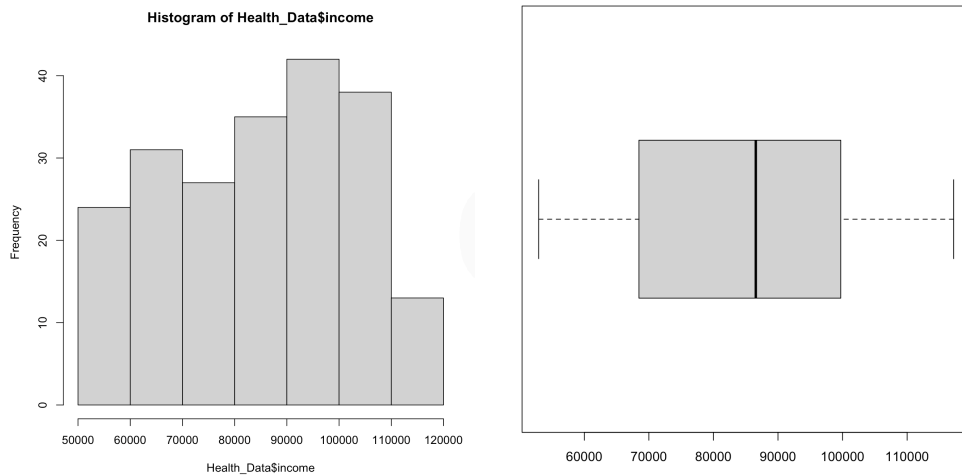
2.1. Central Tendency Measures

Variable	Mean	Median	Mode
sbp	127.7333	123	120
dbp	82.76667	82	74 80 82
income	85194.49	86560.5	No mode
age	26.51429	27	26

2.2. Five-figure summary - Monthly income (income)

Min.	1st Qu.	Median	3rd Qu.	Max.
52933	68636	86560	99696	117210

2.3. Visualisation - Monthly income (income)



The histogram shows a slightly left-skewed distribution. The box plot shows that the median is pretty centred, indicating no strong skewness. The left and right whiskers are of similar length, indicating that the distribution is approximately symmetric. There are also no obvious outliers visible in this group.

3. Hypothesis Testing 1: systolic blood pressure ~ peptic ulcer

The objective of the research is to determine whether there is an association between systolic blood pressure and the presence or absence of a peptic ulcer.

3.1. Variables Characteristic

Systolic blood pressure is a dependent variable measured, whereas peptic ulcer is an independent variable that may influence and explain the overall outcome.

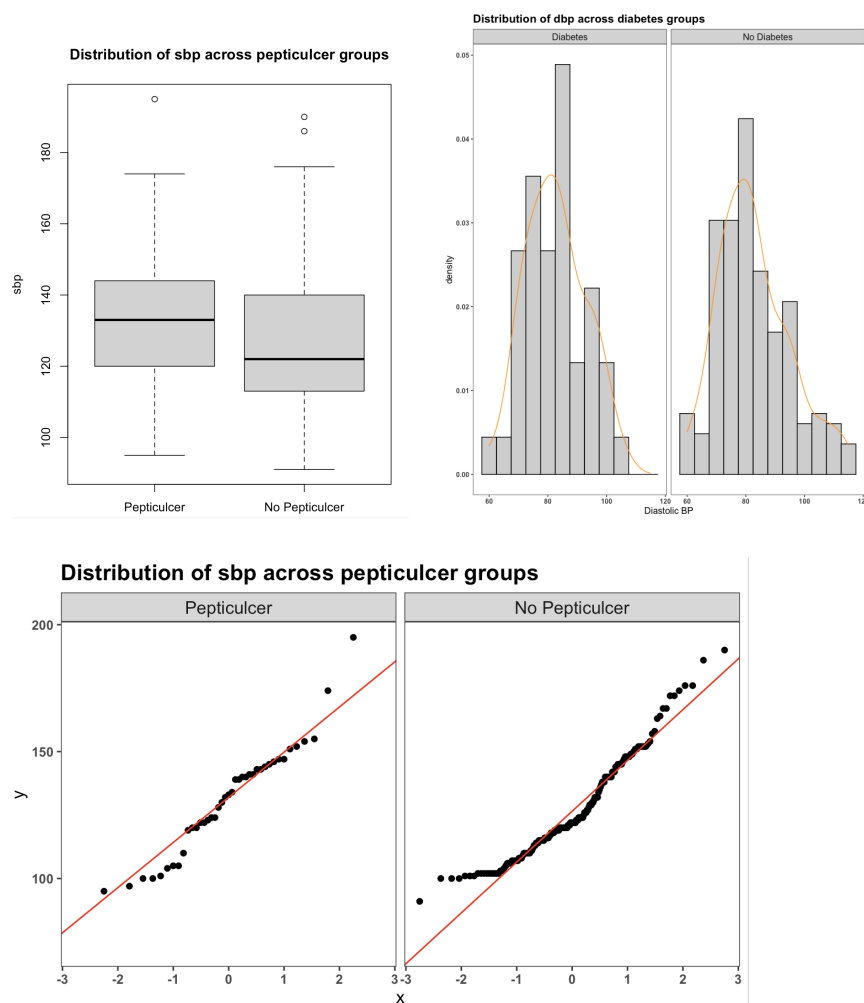
dependent variable (outcome)	systolic blood pressure	numeric variable, continuous scale
Independent variable (predictor)	peptic ulcer	categorical factor variable with two levels ("Pepticulcer "/ "No Pepticulcer ")

The boxplot indicates a few high-value outliers in both peptic ulcer groups (195 for "Pepticulcer", and 186, 190 for "No Pepticulcer"). The whiskers and box widths are similar in length, suggesting that the overall spread of sbp is similar across peptic ulcer groups. There are slightly higher medians and means in the Pepticulcer group, but

overall variability in sbp across the two groups is similar.

Additionally, the histogram with the density curve shows an approximately normal distribution, indicating no substantial differences between the groups.

	Mean	Median	IQR	sd
Pepticulcer	131.32	133	24	21.38
No Pepticulcer	126.86	122	27	19.69



3.2. Normality Assumption (sbp)

The Shapiro-Wilk normality test yielded a p-value of 0.0962 for Pepticulcer (n=41), which is > 0.05 , indicating no evidence of non-normality, whereas the No-Pepticulcer group (n = 169) yielded $p = 2.119e-06$, smaller than 0.05, indicating that the

distribution of sbp in this No Peptic ulcer is different from normal, which is also visible in the QQ plot representation with a curvature away from the straight line, especially in the upper and lower tail.

Shapiro test	P-value	Sample size
Peptic ulcer	$p = 0.0962$	41
No Peptic ulcer	$p = 2.119e-06$	169

3.3. Hypothesis formulation

H0: The RANK SUM of systolic blood pressure (sbp) is the same for patients with and without peptic ulcer.

H1: The RANK SUM of systolic blood pressure (sbp) differs between patients with and without peptic ulcer.

3.4. Non-Parametric Wilcoxon rank-sum test

A non-parametric Wilcoxon rank-sum test was selected, given that the dependent variable (systolic blood pressure) is numeric and the predictor (peptic ulcer) is a binary categorical variable with two independent groups ("Peptic ulcer"/ "No Peptic ulcer"). Assessment of normality indicated that the distribution of systolic blood pressure is non-normal, thereby violating the assumptions required for the parametric test.

3.5. Test Results

The Wilcoxon rank-sum test yielded $p = 0.1434$, which is greater than the significance level $\alpha = 0.05$, so we fail to reject the null hypothesis. The results did not reveal a statistically significant difference in the systolic blood pressure (sbp) between patients with and without a peptic ulcer.

4. Hypothesis Testing 2: diastolic blood pressure ~ diabetic condition

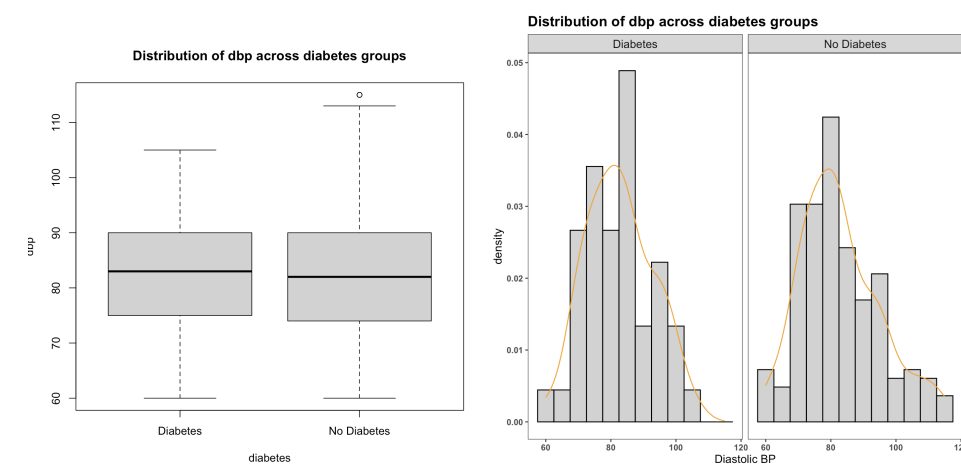
dependent variable (outcome)	diastolic blood pressure	numeric variable, continuous scale
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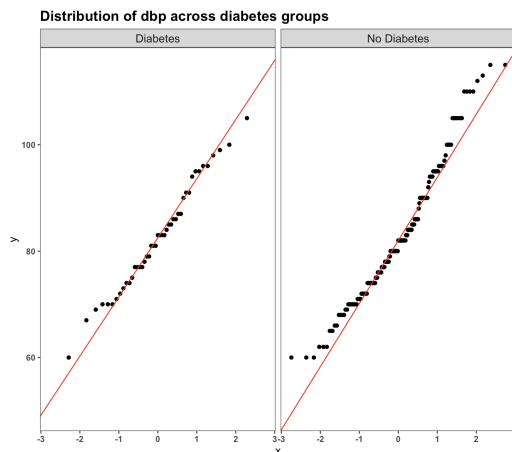
Independent variable (predictor)	diabetes	categorical factor variable with two levels ("Diabetes"/ "No Diabetes")
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4.1. Data Characteristic

	Mean	Median	IQR	sd
Diabetes	82.53	83	15	10.13
No Diabetes	82.83	82	16	12.17

The central tendency and disparity measures indicate that there is similarity in the distribution of dbp across diabetes and no-diabetes. The whiskers and box widths are similar in length, suggesting that the overall spread and variability of dbp are similar between diabetes and no-diabetes groups. The histogram with the density curve shows an approximately symmetric distribution, although there is some right skewness in both groups, with a few outliers around 115 in the No Diabetes group. The QQ plot indicates approximately linear relationships across both groups, with some light curvature, especially in the upper and lower tails.





4.2. Normality Assumption (diastolic blood pressure)

The Shapiro-Wilk normality test yielded a p-value of 0.7797 for Diabetics, which is > 0.05 , indicating no evidence of non-normality, whereas the No-Diabetes group yielded $p = 0.0002865$, smaller than 0.05, indicating that the distribution of dbp in the No Diabetes groups is different from normal, which is also visible in the QQ plot representation with a light curvature in the upper and lower tail.

Shapiro test	P-value	Sample size
diabetic	$p = 0.0962$	45
No diabetic	$p = 2.119e-06$	165

4.3. Hypothesis

H0: The rank sum of diastolic blood pressure (dbp) is the same among diabetic and non-diabetic participants.

H1: The rank sum of diastolic blood pressure (dbp) differs among diabetes and non-diabetes.

4.4. Mann-Whitney U test

A nonparametric Mann-Whitney U test was selected, given that the dependent variable (diastolic blood pressure) is a continuous numeric variable and the predictor (diabetes) is a binary categorical variable with two independent groups ("Yes" / "No") to compare. Assessment of normality indicated that the distribution of diastolic blood pressure

(dbp) violates the assumption of normality, suggesting that a nonparametric test is appropriate.

4.5. Test Results

Mann-Whitney U test Results	
p-value = 0.7999	p-value is the significance level of the t-test

The p-value of the test is 0.7999, which is greater than the significance level $\alpha = 0.05$, so we fail to reject the null hypothesis. The Wilcoxon rank-sum test found no statistically significant difference in diastolic blood pressure between diabetic and non-diabetic groups ($W = 3804.5$, $p\text{-value} = 0.7999$).

However, given the mixed-normality results, a robustness check was performed using **Welch's t-test, which** confirmed the non-significant result ($t = -0.16649$, $df = 82.075$, $p\text{-value} = 0.8682$, 95% CI $[-3.845, 3.251]$). A wide CI spanning both negative and positive values indicates the estimate is imprecise and the effect is negligible. There is no meaningful difference in diastolic blood pressure between diabetic and non-diabetic groups

5. Hypothesis Testing 3: systolic blood pressure ~ occupation

The objective of the hypothesis test is to determine whether the systolic blood pressure (sbp) is different across four occupational group ("*GOVT JOB*", "*PRIVATE JOB*", "*BUSINESS*", "*OTHERS*").

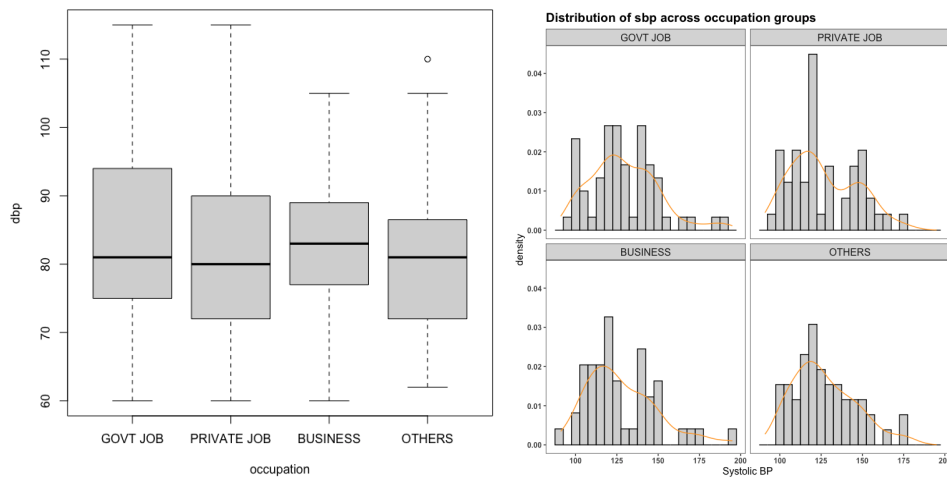
dependent variable (outcome)	systolic blood pressure	numeric variable, continuous scale
Independent variable (predictor)	occupational group	categorical variables with 4 levels (" <i>GOVT JOB</i> ", " <i>PRIVATE JOB</i> ", " <i>BUSINESS</i> ", " <i>OTHERS</i> ")

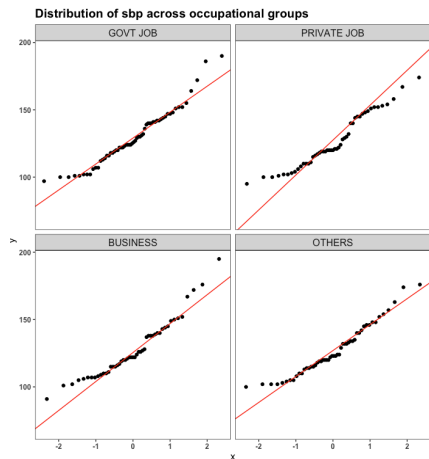
5.1. Data characteristic

	Mean	Median	IQR	sd
GOVT JOB	129.83	125.5	26	20.57
PRIVATE JOB	126.34	120	35	19.89
BUSINESS	127.851	122	29	21.33
OTHERS	127.03	123	26	18.77

The data and boxplot indicate that:

- The median sbp is approximately similar across 1, 2, and 4 occupational groups, whereas the median of group 3 is slightly higher compared to the other three groups
- All 4 groups have the same box sizes (IQR) and pretty similar whisker lengths, suggesting that the variability across the sbp groups is relatively consistent.
- There is only one outlier in group 4 and no obvious outliers in groups 1,2 and 3.
- The histogram with density curve points at a similar approximation of symmetry, with right skewness across all variables, while the QQ plot suggests non-linearity.





5.2. Normality Assumption (systolic blood pressure)

The Shapiro-Wilk normality test yielded p-values < 0.05 for three variables, indicating that the distribution of GOVT JOB, PRIVATE JOB and OTHERS is different from normal, whereas BUSINESS follows a normal distribution.

Shapiro test	P-value
GOVT JOB	$p = 0.01929$
PRIVATE JOB	$p = 0.02096$
BUSINESS	$p = 0.0761$
OTHERS	$p = 0.0213$

5.3. Hypothesis formulation

H0: The rank sum of systolic blood pressure (sbp) is the same across the occupational groups

H1: The rank sum systolic blood pressure (sbp) is different across the occupational groups

5.4. Kruskal-Wallis t-test

A parametric Kruskal-Wallis test was selected, given that the dependent variable (sbp) is a continuous numeric variable and the predictor (occupation) is a categorical variable with four independent groups to compare. Assessment of normality indicated

that the distribution of systolic blood pressure across the occupancy groups does not follow normal distribution (with the exception of one group), suggesting that a non-parametric test is appropriate.

5.5. Test Results

Kruskal-Wallis t-test			
Source	Chi-sq	df	p-value
Occupation	0.77906	3	0.8545

The p-value of the test is 0.8545, which is greater than the significance level $\alpha = 0.05$; hence, we fail to reject the null hypothesis. The test indicated that the systolic blood pressure (sbp) is the same across the occupational groups

6. Correlation between systolic and diastolic BP, including scatter plot

The objective of this analysis is to assess whether there is any association between sbp and dbp?

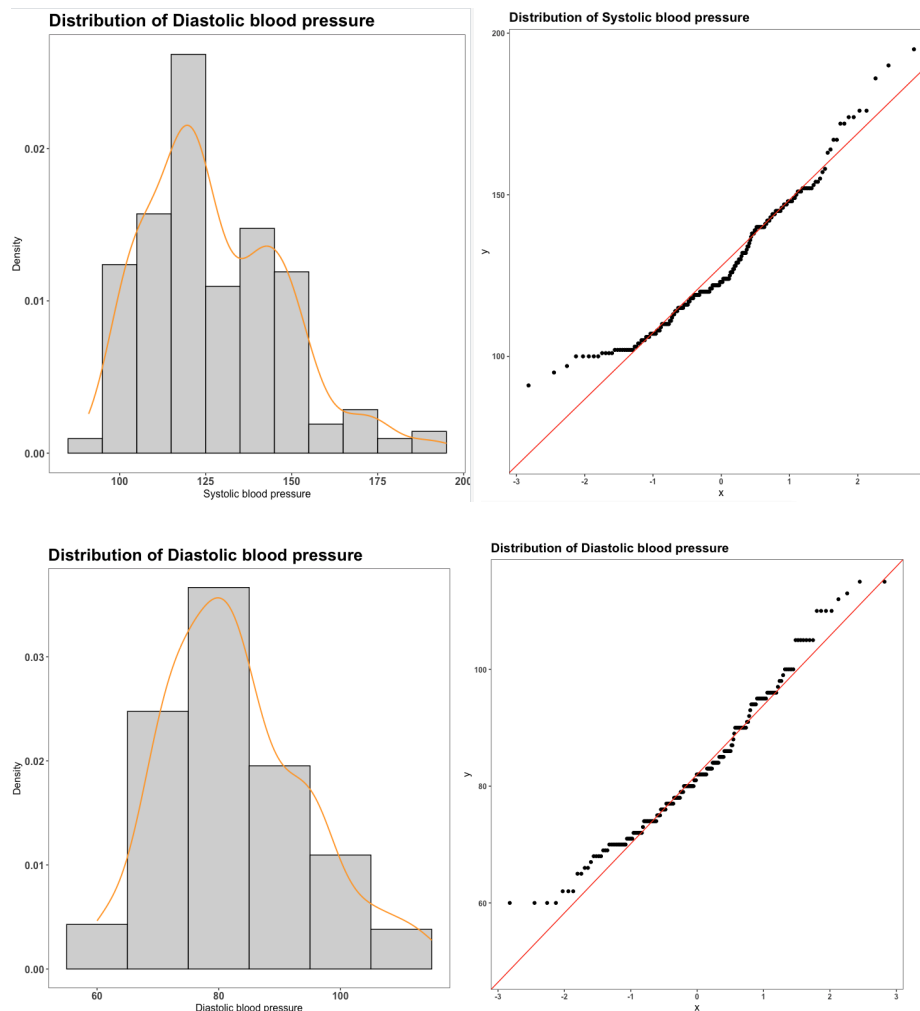
6.1. Normality assumption (Anderson-Darling normality test)

The Anderson-Darling normality test yielded for a sbp $p = 3.319e-06$ and for dbp $p = 0.0001226$, which are lower than the 0.05 significance level; hence, we can reject H_0 and accept H_1 , which tells us that there is evidence that the dbp and sbp distribution deviates from a normal distribution

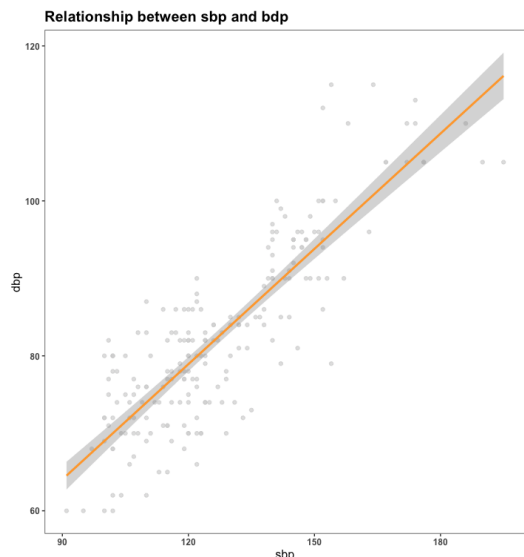
Anderson-Darling test	p-value
sbp	A = 2.4472, p-value = 3.319e-06
dbp	A = 1.8085, p-value = 0.0001226

6.2. Visual inspection

Visual inspection also indicates non-normality across the variables. A light right-skewness is visible on the histogram, and the QQ plot showcases a pronounced curvature away from the straight line, in the upper and lower tails



The scatter plot indicates an approximately linear positive relationship between systolic and diastolic blood pressure. As systolic blood pressure increases, diastolic blood pressure also tends to increase. The points cluster around a straight line without obvious curvature. Although some variability is present (moderate spread) but it does not obscure the overall linear trend. There are a few observations further from the line, but none appear to dominate the relationship.



6.3. Pearson correlation test

The visual assessment of sbp and dbp relationship indicates approximate linearity, and there are only a handful of outliers, suggesting a **Pearson correlation test**.

The following hypotheses are tested:

H0: There is no linear correlation between sbp and dbp

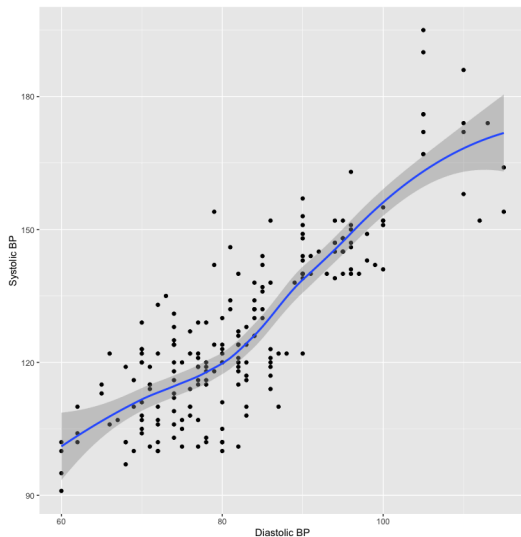
H1: There is a linear correlation between sbp and dbp

Pearson correlation test yielded $p < 0.001$ and $\text{cor } 0.846808$. The 95% confidence interval (0.804–0.881) indicates that the true population correlation is likely to lie within this range. As the interval is narrow and it does not include zero, the correlation is statistically significant.

Pearson correlation test	p-value
t	t = 22.961
df	208
p-value	p < 2.2e-16
95% confidence interval:	0.8035436, 0.8811723
cor 0.846808	0.846808

7. Simple linear regression for predicting the diastolic BP by systolic BP

The scatter plot indicates a linear relationship between dpb and sdp. The Correlation analysis confirms a strong positive relationship (0.846808) between the variables.



8. Population regression equation to predict the diastolic BP by systolic BP.

- Dependent (response) variable: **dbp**
- Independent (predictor) variable: **sbp**

Regression Equation: $dbp = 19.407 + 0.496 \cdot (sbp)$

Slope 0.496: How much dbp is expected to change for each 1 unit increase in SBP

The model summary indicated that both p-values for the intercept and the predictor variable are highly significant ($p < 0.001$). So, we can reject the null hypothesis and accept the alternative hypothesis. This means that there is a significant association between the predictor and the outcome variables.

- The regression model is **significant** ($F(1,208) = 527.2, p < 0.001$)
- The model **explains 71.5% of of the variation** in dbp is explained by sbp
- Spb is a **significant predictor** of dbp ($\beta = 0.4960, t = 22.96, p < 0.001, 95\% \text{ CI } [0.45, 0.53]$). One unit increase in sbp is associated with a 0.50 increase in dbp, with the narrow CI confirming a precise and reliable estimate.

9. Reflection

This exercise contributed to very valuable learnings and observations:

- Applying a framework for test selection is critical. As Ranganathan (2021) argues, selecting the appropriate test depends on first focusing on descriptive statistics and understanding the research question. Analysing the data's characteristics should precede inferential analysis, during which hypotheses are formulated, and assumptions about normality, outliers, and distribution are validated. This sequence guides the choice of the most suitable test.
- Knowing which variable is being discussed and measured (the outcome) and which one may have an effect (the predictor) is critical for hypothesis formulation. Moreover, variable types drive test selection. It's important to understand the nature of the variables and how they are stored in R. While working with the health dataset, I realised that variables may be stored as numerical labelled values (num, dbl) while actually representing categories (e.g., Yes/No or "GOVT JOB," "PRIVATE JOB," "BUSINESS," "OTHERS"). This requires careful inspection before analysis, as the statistical meaning might otherwise be changed. Numerical variables can serve as outcomes in a parametric test, while the categorical variables define the groups being compared. Understanding this distinction proved critical for selecting appropriate tests and correctly interpreting results.
- Initially, I focused on applying statistical tests, but the exercise highlighted the importance of data preparation and variable interpretation. Converting labelled variables, such as diabetes and occupation, into factors ensured that R treated them as categorical rather than numerical, leading to more appropriate analyses.
- Assumptions must be checked before testing. Before conducting any test, it is essential to assess assumptions about normality, outliers, distribution, and linearity using formal tests and visual inspections, such as boxplots, histograms with density curves, and Q-Q plots. This step highlights that understanding the data first is critical for choosing an appropriate statistical method.
- Hypothesis formulation leads to test selection. I initially focused on formulating a hypothesis and choosing a test, but I have learned that, while the research question stays the same, the hypothesis depends on the test's assumptions

and the nature of the variables being compared. This distinction helps decide whether it is more appropriate to compare group means with parametric tests or rank sums/differences with nonparametric tests. It highlights the importance of choosing statistical methods that align with the data's underlying assumptions.

- Non-significant results do not prove that groups are identical; they simply indicate insufficient evidence to conclude a difference exists. This distinction is key to interpreting hypothesis-testing results.
- Statistical significance is very different from practical or clinical importance. The P-value alone indicates that the correlation is real, not due to chance, but it does not indicate the direction, strength, or precision, nor does it account for the small sample size. The CI matters because it shows that the correlation is not just "significantly different"; it's a strong, tightly bound relationship. Even at the lower bound, it is still a strong positive correlation, not a weak one that happened to reach the significance bar. CI is doing all the real interpretive work.
- Visuals support interpretation. The box plot, Histogram with density curve, and Q-Q plot provided useful information for identifying potential outliers and comparing variability between groups, generating an initial image and understanding of the data. However, these visuals do not constitute decisive evidence; they must be supported by statistical testing.
- Careful data management is key. Data preparation, reproducibility and proper variable encoding are essential, and debugging is a critical practice that supports them. During the analysis, I discovered that repeatedly converting labelled variables could inadvertently overwrite data and introduce missing values. This highlighted the value of maintaining an untouched copy of the original dataset, performing all variable transformations in a dedicated data preparation section, and structuring the script so that the entire analysis can be reproduced consistently within a trustworthy workflow.
- Finally, although less pronounced in this exercise, as Schober, Bossers and Schwarte (2018) argue, it is important to be mindful that decisions should not be guided solely by p-values. The overall results—including the effect size (how large the effect is) and the confidence interval (how precise the estimate is)—help determine whether the effect is actually meaningful in practice.

This exercise has reinforced the importance of understanding different types of variables, checking statistical assumptions, and then selecting the test that aligns with the research question and data structure. Moreover, it highlights the value of applying a framework that guides data evaluation and preparation, visual exploration, statistical testing, and interpretation when drawing final conclusions.

References

Bader, M. and Leuzinger, S. (2025) *R-ticulate: A Beginner's Guide to Data Analysis for Natural Scientists*. 1st edn. Hoboken: Wiley.

Ranganathan, P. (2021) 'An introduction to statistics: choosing the correct statistical test', *Indian Journal of Critical Care Medicine*, 25(Suppl 2), pp. S184–S186. DOI: 10.5005/jp-journals-10071-23815

Schober, P. S., Bossers, S. M. and Schwarte, L. A. (2018) 'Statistical significance versus clinical importance of observed effect sizes: what do P values and confidence intervals really represent?', *Anesthesia & Analgesia*, 126(3), pp. 1068–1072. DOI: 10.1213/ANE.0000000000002798

University of Essex (2026) *Health Survey Dataset* [dataset]. Numerical Analysis Module. Unpublished teaching dataset.